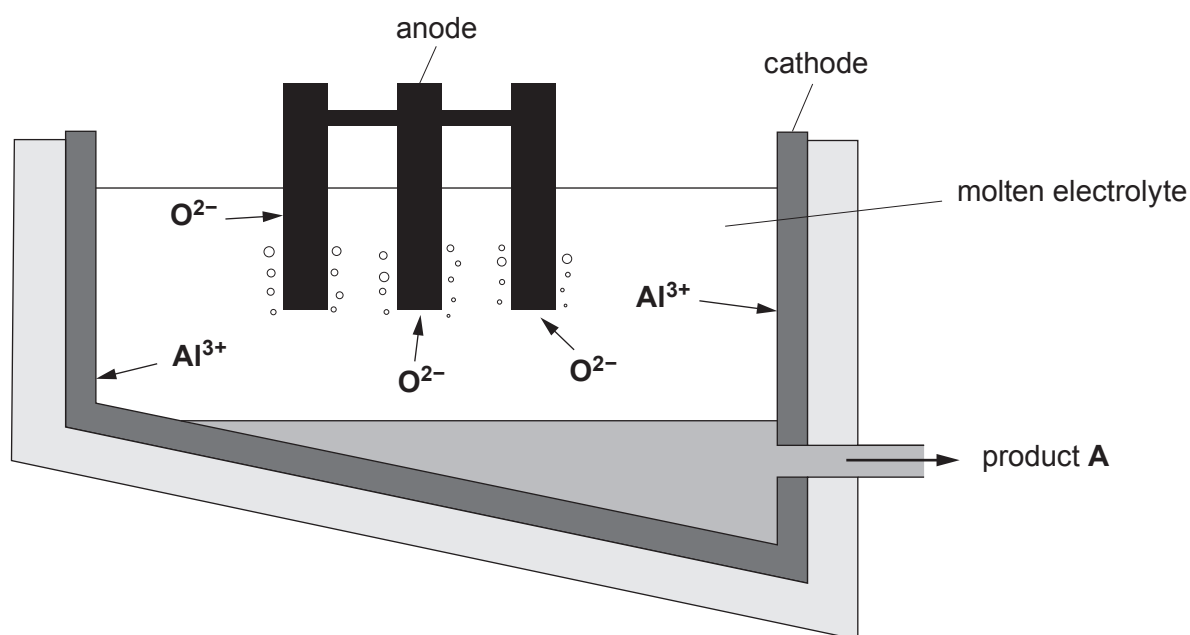


2. The diagram shows an electrolysis cell used in the extraction of aluminium from alumina.



bauxite	electrical	molten alumina	positive
negative	cryolite	molten aluminium	oxygen
			light

(a) Choose words from the box to complete the following sentences. [5]

The molten electrolyte is a mixture of alumina and

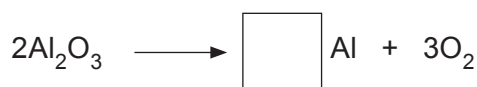
Product **A** is

Oxygen is formed at the anode which is the electrode.

Alumina is obtained from an ore called

The process of extracting aluminium is expensive because it uses a lot of energy.

(b) Balance the symbol equation that shows the overall reaction. [1]

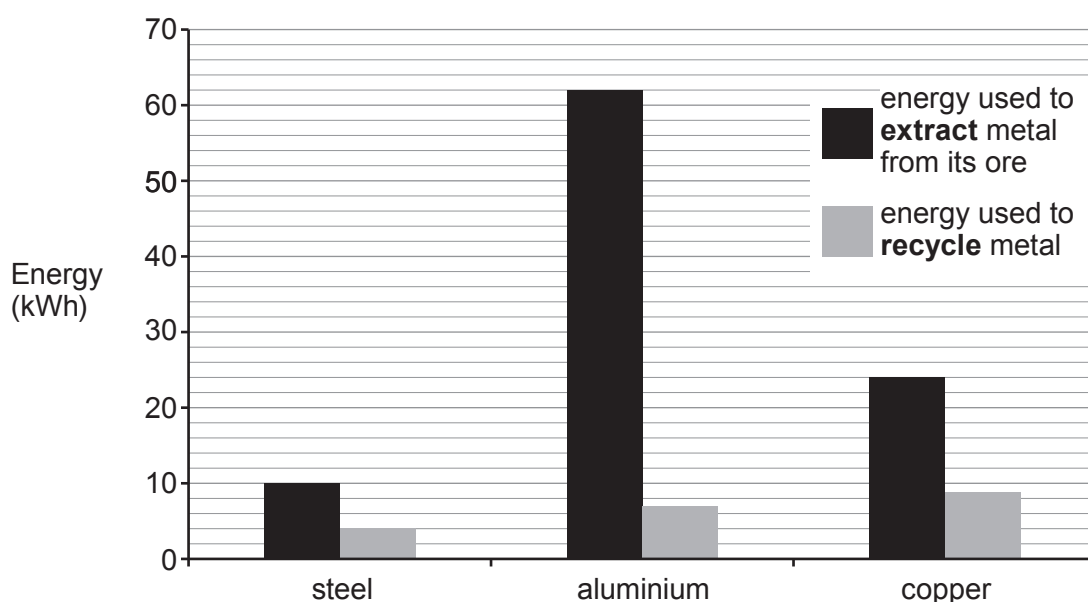


- (c) 250 tonnes of an aluminium ore can produce 195 tonnes of alumina.
Calculate the percentage of alumina in the ore.

[2]

Percentage = %

- (d) The chart compares the amount of energy used to extract and to recycle three metals.



Use the information in the chart to answer the following question.

Put a tick (✓) in the box next to the statement that describes why it is more cost effective to recycle aluminium than steel and copper. [1]

The energy used to extract metals is greater than that used in recycling them

☐

The difference between the energy used to extract and the energy used to recycle is the greatest

☐

The energy used in recycling is less than for copper but greater than for steel

☐
